Notes: Beraja, Fuster, Hurst, and Vavra (QJE, 2019)

Regional Heterogeneity and the Refinancing Channel of Monetary Policy

Contents

1 Big picture

- **Question.** How much does the *regional distribution of housing equity* shape the refinancing channel of monetary policy? The paper asks whether the same fall in mortgage rates can have very different aggregate and local effects depending on where home equity is concentrated.

- **Central idea.** Expansionary monetary policy works partly by lowering mortgage rates, which encourages borrowers to refinance and sometimes extract equity to finance current spending. That channel is strong only when households have enough equity to refinance.
- **Main conceptual contribution.** The paper moves the focus from average national housing conditions to the *cross-regional distribution* of housing equity. The key claim is that time variation in this distribution changes both the strength and the geography of monetary stimulus.
- **Empirical setting.** The core empirical episode is the first round of quantitative easing (QE1) in late 2008. The paper uses detailed mortgage microdata to compare metropolitan areas with high and low pre-QE housing equity.
- **Main empirical findings.**
  - Refinancing surged after QE1, but it surged far more in places where homeowners had more equity.
  - Cash-out refinancing rose much more in high-equity regions.
  - Spending, measured by new car purchases, rose more in the same high-equity regions.
  - In the hardest hit regions, where many borrowers were underwater, the refinancing channel was weak.
- **Time-variation result.** The 2001 recession looked very different from the Great Recession. House prices were rising rather than collapsing, regional house price growth was much less tied to local labor market distress, and refinancing was much stronger overall. This suggests that the effects of monetary policy through refinancing are not stable over time.
- **Why regional evidence alone is not enough.** The cross-regional empirical design shows who benefits more from lower rates, but it differences out common aggregate effects. To translate regional evidence into aggregate implications, the authors build a heterogeneous-household model with mortgage borrowers and lenders.
- **Main model result.** Under 2008 conditions, a fall in mortgage rates raises aggregate consumption only modestly and mainly benefits better-performing regions, so monetary easing increases cross-regional consumption inequality. Under a 2001-style equity distribution, the same rate cut produces much larger aggregate stimulus and can even mildly reduce regional inequality.
- **Economic mechanism.** Underwater households cannot refinance without putting in cash. That creates a highly nonlinear relation between equity and refinancing. Because the equity distribution shifts over time, the economy's response to the same interest-rate change also shifts over time.
- **Policy implication.** When many borrowers are underwater, standard monetary easing is weak exactly where local conditions are worst. Mortgage-market interventions that relax refinancing constraints or reduce debt can complement monetary policy and reduce the trade-off between stimulus and regional inequality.
- **Bottom line.** The paper's message is that central banks should track not only average leverage or average house prices, but the *distribution of collateral across space*. That distribution is an important state variable for monetary transmission.

2 Data and measurement (key objects)

2.1 Why this paper studies the U.S. mortgage market

The paper focuses on the mortgage market because housing collateral is central to household borrowing in the United States. When mortgage rates fall, households can refinance, reduce monthly payments, and sometimes cash out housing equity. This is the "refinancing channel" of monetary policy. Because housing markets are locally segmented, regional house price shocks generate regional differences in home equity and therefore in the strength of that channel.

2.2 Main data sources

CRISM data. The main microdata come from Equifax's Credit Risk Insight Servicing McDash (CRISM) data set, which merges Black Knight mortgage-servicing data with Equifax credit records.

- Coverage starts in 2005.
- The data cover roughly two-thirds of the U.S. mortgage market.
- The linked structure allows the authors to observe when a borrower refinances, the size of the old mortgage, the size of the new mortgage, and therefore the amount of equity extracted in the refinancing process.

HMDA data. The paper also uses Home Mortgage Disclosure Act (HMDA) data.

- HMDA has broader coverage and a longer time series.
- It is especially useful for extending the analysis back to the 2001 recession.
- Its limitation is that it does not contain outstanding loan balances, so it cannot directly measure borrower equity or cash-out amounts.

House prices and labor market data. House price indexes come primarily from CoreLogic. Local labor market conditions are measured using unemployment changes and local income data. Car purchases are measured using Polk data on new car purchases at the MSA level.

2.3 Equity measurement

The paper defines borrower home equity as one minus the combined loan-to-value ratio:

$$\text{Equity}_{hjt} = 1 - \text{CLTV}_{hjt}. \tag{1}$$

For each household, combined mortgage balances include the first mortgage plus any second liens, divided by estimated current house value.

Current property values are constructed by taking appraisal values at loan origination and updating them using location-specific house price indexes. The preferred local summary statistic is the **equity of the median borrower** in location j at time t, denoted $E^{\text{med}}_{j,t}$.

The paper places special emphasis on:

$$E^{\text{med}}_{j,\text{Nov 2008}}, \tag{2}$$

which captures the local equity distribution immediately before QE1.

Interpretation. This variable is not just a local wealth measure. It is the local state variable that governs whether the representative borrower in that place can refinance and how much equity can be extracted.

2.4 Refinancing and participation outcomes

The main mortgage activity outcome is the **monthly refinancing propensity** in an MSA: the dollar amount of refinance originations in a month divided by outstanding mortgage balances in the prior month.

The paper also studies:

- monthly cash-out refinancing volumes,
- cumulative cash extracted through refinancing,
- and car-purchase responses following refinancing.

At the household level, the authors identify a new car purchase by looking for a sufficiently large increase in the household's auto loan balance from one month to the next.

2.5 Why QE1 is a useful event

QE1 is useful because it generated a sharp decline in mortgage rates. Mortgage applications responded immediately, and originations rose with the normal lag induced by underwriting and processing. This episode therefore offers a relatively clean way to study how lower rates pass through to refinancing and spending.

The paper emphasizes that the increase in originations after QE1 was mostly a refinance boom, not a purchase-mortgage boom. That matters because the paper is about the response of existing borrowers.

2.6 Key descriptive facts

Several descriptive numbers are economically important:

- Over the first half of 2009, the median old mortgage rate among refinancers was 6.125%, while the median new rate was 4.875%.
- For the average refinanced first-lien balance of about $206,000, keeping the balance unchanged implies a monthly payment decline of at least $160.
- Discounted over about seven years, the implied present value of pretax savings is roughly $11,400 using payment savings and roughly $15,000 using interest savings.
- Over 2009:H1, the mean equity withdrawal in the data is about $25,000 and the median is about $7,400.

Interpretation. These are first-order changes for households. So if refinancing is blocked in some regions, the resulting reduction in stimulus is potentially large.

2.7 What Figure II shows about equity distributions

Before the nationwide house price collapse, the equity distributions of borrowers across places like Chicago, Las Vegas, Miami, Philadelphia, and Seattle looked broadly similar. By November 2008 they looked radically different.

Examples emphasized by the paper:

- In Las Vegas the median borrower had negative equity, around -0.17.
- In Miami the median borrower had about zero equity.
- In Seattle and Philadelphia the median borrower still had roughly 0.25–0.30 equity.

In Miami roughly half of borrowers had negative equity by November 2008, while in Las Vegas it was around 70%. In Philadelphia and Seattle the corresponding share was only about 6–10%.

Interpretation. The refinancing channel was operating on two very different borrower balance-sheet distributions across regions at the same point in time.

3 Models and empirical specifications (all equations + interpretation)

3.1 Economic mechanism and empirical hypotheses

The paper's mechanism has two parts.

- **Collateral constraint.** Lenders generally require a minimum amount of equity for refinancing, even if the borrower is not extracting cash.
- **Cash-out margin.** Conditional on refinancing, the amount of equity that can be withdrawn depends directly on how much equity the household already has.

This yields two empirical predictions:

1. Lower mortgage rates should induce more refinancing and more spending where equity is high.
2. When the regional distribution of equity changes over time, the aggregate consequences of monetary easing should also change over time.

3.2 Baseline QE1 difference-in-differences specification

To estimate how local equity affects the refinancing response to QE1, the paper runs:

$$\text{Refi}_{j,t} = \alpha_j + \alpha_t + \beta(E^{\text{med}}_{j,\text{Nov 2008}} \times \text{postQE}) + \Theta(X_{j,\text{Nov 2008}} \times \text{postQE}) + \varepsilon_{j,t}. \tag{1}$$

Here:

- $\text{Refi}_{j,t}$ is the monthly refinancing propensity in MSA j,
- α_j are MSA fixed effects,

- α_t are month fixed effects,
- postQE equals 1 for the six months after the QE1 announcement,
- and $X_{j,\text{Nov 2008}}$ is a rich vector of local controls.

The post-QE period starts in February 2009 rather than December 2008 because refinancing originations occur with a lag after applications are filed.

Interpretation of β. A positive β means regions with higher pre-QE equity experienced a larger refinancing increase after QE1. Since the regression includes MSA fixed effects, it is not using permanent level differences across places. Since it includes time fixed effects, it is not using national refinancing waves. It is using the interaction between local pre-QE balance-sheet conditions and the QE1 shock.

3.3 Cross-regional comovement of house prices and unemployment

To document how housing shocks line up with local economic distress over time, the paper estimates:

$$\Delta \log HP_{i,t} = \beta_t \, \Delta UR_{i,t} + \gamma_t + \zeta_i + \varepsilon_{i,t}. \tag{2}$$

Here $\Delta \log HP_{i,t}$ is annual state-level house price growth and $\Delta UR_{i,t}$ is the annual change in unemployment. The coefficient β_t varies by year.

Interpretation. This equation is descriptive but important. It shows whether the places with the worst labor markets are also the places where collateral is being destroyed. In the Great Recession the answer is yes; in 2001 the answer is essentially no.

3.4 Aggregate refinancing equations across time

To connect regional house price distributions to aggregate refinancing, the paper estimates two related time-series regressions:

$$\text{Refi}_t = \omega_0 + \omega_1 \text{Rate}_t + \omega_2 \Delta HP_t + \omega_3 (\text{Rate}_t \times \Delta HP_t) + \omega_4 SD(\Delta HP_t) + \omega_5 (\text{Rate}_t \times SD(\Delta HP_t)) + \varepsilon_t, \tag{3}$$

$$\Delta\text{Refi}_t = \omega_0 + \omega_1 \Delta\text{Rate}_t + \omega_2 \Delta HP_t + \omega_3 (\Delta\text{Rate}_t \times \Delta HP_t) + \omega_4 SD(\Delta HP_t) + \omega_5 (\Delta\text{Rate}_t \times SD(\Delta HP_t)) + \varepsilon_t. \tag{4}$$

Here:

- Refi_t is aggregate refinancing activity,
- Rate_t is either the mortgage refinancing incentive, the 12-month change in mortgage rates, or monetary policy surprises,
- ΔHP_t is the cross-state mean of annual house price growth,
- $SD(\Delta HP_t)$ is the cross-state dispersion of annual house price growth.

Interpretation. The interaction terms are the heart of the paper. They ask whether a rate decline translates into more refinancing precisely when average house price growth is higher and when the cross-regional equity distribution is more favorable for refinancing.

3.5 Household model: environment

The structural model is an incomplete-markets model with mortgage borrowers and lenders.

Income process.

$$\log y_t^{ij} = \mu_y^j + \log y_{t-1}^{ij} + \varepsilon_t^{ij}. \tag{3}$$

House price process.

$$\log q_t^j = \mu_q^j + \log q_{t-1}^j + \nu_t^j. \tag{4}$$

Households live in regions $j = 1, 2, \ldots, J$, receive stochastic income, and own one unit of housing. House prices follow region-specific random walks, so a region's recent housing shock directly changes local collateral values.

3.6 Mortgage structure and refinancing

The model uses fixed-rate mortgages with infinite maturity for tractability. If the household last refinanced at time τ_0, it pays a fixed nominal mortgage payment each period. Refinancing is allowed at any later date $\tau > \tau_0$ by paying a fixed cost proportional to the current house price.

When a household refinances, it resets the mortgage rate to the current market mortgage rate and borrows up to the maximum loan-to-value ratio γ.

3.7 Budget constraints with and without refinancing

If the household does *not* refinance in period t, its sequential budget constraint is:

$$p_t c_t + a_{t+1} \leq a_t(1 + r_t) + y_t - \gamma r_{\tau_0}^m q_{\tau_0}. \tag{5}$$

If the household *does* refinance in period t, its budget constraint becomes:

$$p_t c_t + a_{t+1} \leq a_t(1 + r_t) + y_t - \gamma r_t^m q_t + \gamma(q_t - q_{\tau_0}) - F_t q_t. \tag{6}$$

The term $\gamma(q_t - q_{\tau_0})$ is the equity extraction term, while $F_t q_t$ is the fixed refinancing cost.

Interpretation. Refinancing has two benefits: locking in a lower rate and extracting accumulated equity. Its cost is the fixed refinancing cost plus the fact that future mortgage balances may become larger after cash-out.

3.8 Recursive value function

The household compares the value of refinancing and not refinancing:

$$V(\cdot) = \max\{V^{\text{norefi}}(\cdot),\ V^{\text{refi}}(\cdot)\}. \tag{7}$$

The paper normalizes the state variables by current house prices so that the problem is stationary. The important economic implication is that the optimal policy is characterized by a threshold in extractable equity: when equity is too low, households do not refinance; once equity is high enough, refinancing becomes optimal.

3.9 Why the policy function is nonlinear

The refinancing decision follows a threshold rule because of the fixed refinancing cost. Households wait in an inaction region and refinance only when the state moves enough to justify paying that cost.

The paper's most important theoretical point is that the refinancing probability is *highly nonlinear* in extractable equity. Underwater households do not respond at all, mildly positive-equity households respond somewhat, and high-equity households respond a lot.

Interpretation. This nonlinearity is exactly why the *distribution* of equity matters and not just the mean.

3.10 Role of lenders and aggregation

Mortgage payments are not lost; they are received by lenders. The model therefore includes a representative domestic lender plus foreign lenders. A share θ of mortgage debt is held domestically and $1 - \theta$ is held abroad.

This matters for aggregation. Borrowers consume more when refinancing accelerates cash-out and lowers payments, but domestic lenders consume less when the present value of mortgage receipts falls. Cross-regional regressions difference out these lender offsets; the model is needed to put them back in.

3.11 Calibration

The baseline calibration targets 2008 conditions. Key values are:

- CRRA coefficient $\sigma = 2$,
- discount factor $\beta = 0.95$,
- nominal risk-free rate $r = 0.03$,
- inflation $\pi = 0.02$,
- loan-to-value cap $\gamma = 0.8$,
- domestic lender share $\theta = 0.5$,
- mortgage rate falling from 0.06 to 0.05.

The model is initialized with the aggregate 12.5% house price decline observed in 2008, plus additional region-specific house price and income shocks chosen to mimic the empirical 2008 cross-regional environment. The stochastic refinancing cost process is calibrated so that the model matches the cross-regional QE1 refinancing responses observed in the data.

3.12 Main theoretical predictions

The model predicts:

- regions with more equity refinance more after a rate cut,
- those same regions increase spending more,
- the aggregate consumption response to a rate cut depends on the full equity distribution,
- and the correlation between income and equity determines whether lower rates increase or decrease cross-regional inequality.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the baseline QE1 estimate

The baseline estimate comes from comparing places with different *pre-QE equity levels* around the same national shock. The identifying variation is local balance-sheet heterogeneity at the moment QE1 arrives.

Why this is credible:

- MSA fixed effects absorb permanent local differences.
- Month fixed effects absorb common national refinancing waves.
- The key regressor is predetermined local equity measured in November 2008, just before the refinancing boom is realized.
- The pre-QE trends in refinancing for high- and low-equity MSAs are nearly parallel.

4.2 Why reverse causality is weak

A natural concern is that refinancing itself could affect local conditions. The paper argues that this is limited in the core event study because the movement in mortgage rates is driven by a national policy shock, not by local refinancing behavior.

In the aggregate time-series analysis, the authors go further and use monetary policy surprises from Gertler and Karadi as an alternative rate measure. If anything, reverse causality would bias against their result because stronger refinancing and demand would tend to raise rates, not lower them.

4.3 Rich controls and the role of fixed effects

Table I shows that the local-equity interaction remains positive and statistically significant after controlling for many local factors interacted with the post-QE period:

- unemployment changes,
- income changes,

- FICO scores,
- current mortgage rates,
- loan age,
- jumbo share,
- average mortgage balance,
- ARM share,
- GSE share,
- private securitization share,
- age, education, and homeownership controls.

The coefficient on $E^{\text{med}}_{j,\text{Nov 2008}} \times \text{postQE}$ ranges from about 1.711 in the baseline to 0.483 in the fully saturated specification. The fact that it stays positive and highly significant even after very demanding controls is one of the paper's strongest credibility results.

4.4 Why the result is not just local economic distress

High-equity places also had lower unemployment increases. So perhaps the refinancing result just says that stronger local economies refinance more. The paper addresses this in two ways.

1. It directly controls for unemployment changes and local income changes interacted with post-QE.
2. It shows that the correlation between house prices and unemployment varies dramatically over time. In 2001 there was little correlation, and refinancing patterns reversed. That time variation lines up with the collateral channel, not with a simple recession-severity story.

4.5 Why the result is not just FICO or credit supply composition

Borrowers in high-equity places also tended to have stronger credit characteristics. The authors therefore control for FICO scores and the local composition of mortgages by type.

These controls matter quantitatively, but they do not eliminate the equity effect. The paper interprets this as evidence that borrower credit quality is part of the story but not the whole story. Home equity itself remains a binding state variable for refinancing.

4.6 Local stocks, wealth shocks, and household purchase stories

A broader concern is that rising local wealth or local stock-market gains could independently raise spending and even participation in mortgage products. The paper argues against these explanations by showing that:

- the refinancing patterns line up tightly with measured housing equity,
- the spending response is strongest among borrowers who actually refinance,

- and within refinancers the spending response is larger for *cash-out* refinancers than for non-cash-out refinancers.

That within-refinancer evidence is hard to reconcile with a pure local-wealth story.

4.7 Household-level spending evidence as a mechanism check

The household-level event study is not the source of causal identification, but it is a powerful mechanism check. Car-purchase propensities jump after refinancing and jump almost twice as much after cash-out refinancing as after non-cash-out refinancing.

Interpretation. This makes the mechanism concrete: lower rates matter because they change actual household balance sheets and liquidity, not because they merely proxy for optimism.

4.8 What the paper can and cannot distinguish

The paper is transparent about what the design does not identify directly.

- The regional event study identifies *relative* borrower responses across places, not the full aggregate effect of QE1.
- It cannot directly measure equilibrium consumption offsets from lenders.
- It does not separately identify every possible general-equilibrium channel. Instead, the model focuses on the lender-consumption channel as the main omitted aggregate force.

4.9 Relation to the literature

The paper sits between two literatures.

- It is related to the household debt and housing-collateral literature, but emphasizes how the *distribution* of housing equity shapes monetary transmission.
- It is also related to the heterogeneous-agent monetary transmission literature, but adds realistic mortgage refinancing and cash-out decisions rather than a one-period borrowing technology.

Its distinctive identification contribution is to use regional heterogeneity in collateral at the time of a national policy shock.

5 Tables and figures

5.1 Figure I: mortgage refinancing activity in the United States over 2000–2012

Read:

- The figure plots the Mortgage Bankers Association refinance index and the 30-year fixed mortgage rate relative to its five-year moving average.

- Refinancing is strongly negatively related to mortgage rates.
- QE1 is associated with a sharp mortgage-rate decline and a large refinance boom.

Economic message. Lower mortgage rates can trigger a large refinancing response, but the rest of the paper shows that this response is distributed very unevenly across space.

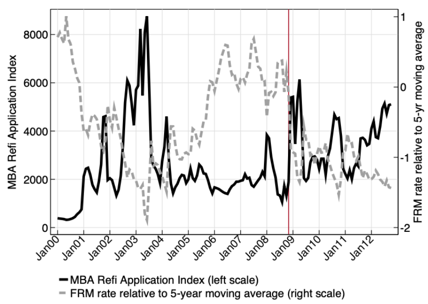

FIGURE I

Mortgage-refinancing Activity in the United States over 2000–2012

Figure shows monthly average of Mortgage Bankers Association (MBA) Refinancing Index (seasonally adjusted; March 1990 = 100) and the 30-year fixed-rate mortgage rate (relative to five-year moving average), also from MBA.

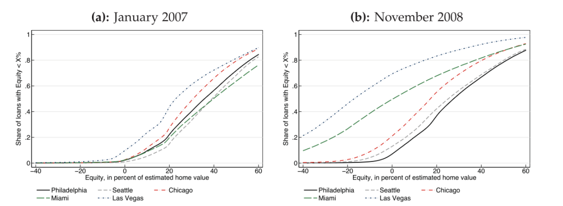

FIGURE II

Distributions of Borrowers' Equity in Their Homes across Five MSAs

Figure shows the cumulative distribution of borrower equity in five illustrative MSAs in January 2007 and November 2008. Equity is measured for each household using CRISM data as the estimated current house value minus total current mortgage debt, divided by estimated current house value (i.e., equity = 1 − CLTV). Distributions are weighted by the mortgage balance.

5.2 Figure II: local equity distributions before and after the crash

Read:

- In January 2007 the equity distributions of selected MSAs looked fairly similar.
- By November 2008 the distributions had diverged dramatically.
- Las Vegas and Miami shifted sharply left; Seattle and Philadelphia remained much healthier.

Economic message. The Great Recession did not produce one national collateral state. It produced very different local collateral states.

5.3 Figure III and Table I: the baseline QE1 refinancing result

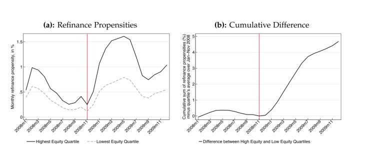

FIGURE III

Mortgage Refinance Activity 2008–2009 in Top and Bottom Quartile of MSAs Defined by Median Borrower Equity in November 2008

Panel A shows monthly refinance propensities in CRISM, defined as the dollar amount of refinance mortgage originations divided by outstanding mortgage amounts in the prior month. Calculations are done over MSA quartile groups for the highest and lowest $E^{med}_{j,Nov2008}$ quartiles. Panel B shows the cumulative difference between the two groups, after subtracting each group's average refinancing propensity from January to November 2008. Vertical lines show the month of the QE1 announcement (November 2008).

TABLE I

THE RESPONSIVENESS OF LOCAL REFINANCING ACTIVITY TO EQUITY AND OTHER LOCAL CHARACTERISTICS AROUND QE1

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$E^{med}_{Nov2008} \times postQE$	1.711***	1.161***	1.530***	0.920***	1.088***	1.856***	1.694***	0.483***
	(0.232)	(0.258)	(0.288)	(0.237)	(0.115)	(0.233)	(0.186)	(0.184)
$\Delta UR_{Jan07-Nov08} \times postQE$		−0.0911***						−0.0511***
		(0.0188)						(0.0131)
$\Delta Income_{2008-09} \times postQE$			2.123*					−1.028
			(1.233)					(0.776)
FICO × *postQE*				0.0107***				0.00876***
				(0.00101)				(0.00180)
Current int. rate × *postQE*					−1.688***			−0.812***
					(0.117)			(0.210)
Loan age × *postQE*					−0.0229***			−0.0326***
					(0.00369)			(0.00521)
Jumbo share × *postQE*					−1.843***			−1.238***
					(0.293)			(0.409)
Average balance (nonjumbo) × *postQE*					−0.218***			−0.231***
					(0.0315)			(0.0881)
ARM share × *postQE*						2.269***		−0.188
						(0.496)		(0.393)
GSE share × *postQE*						0.954***		−0.305*
						(0.240)		(0.170)
Private sec. share × *postQE*						−2.112***		0.190
						(0.616)		(0.534)

TABLE I

(CONTINUED).

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Demographics × *postQE*	N	N	N	N	N	N	Y	Y
MSA fixed effects	Y	Y	Y	Y	Y	Y	Y	Y
Month fixed effects	Y	Y	Y	Y	Y	Y	Y	Y
Adj. R^2	0.87	0.87	0.87	0.88	0.90	0.88	0.89	0.90
Adj. R^2 (within)	0.18	0.22	0.19	0.28	0.35	0.25	0.30	0.38
Observations	4,572	4,572	4,572	4,572	4,572	4,572	4,572	4,572

Notes. Table shows results from regressions of monthly refinancing propensities in 381 MSAs from August 2008 to July 2009, measured in CRISM, on median equity in each MSA interacted with a dummy "postQE" that equals 1 for the period February–July 2009, and other variables also interacted with postQE. Demographic controls in columns (7) and (8) are location-specific shares of individuals with different years of education: 12,13–15,16,>16 (<12 is omitted), shares of individuals with ages: 31–45, 46–60, >61 (21–30 is omitted), homeowner shares, and the share of households with mortgages, from the ACS. Each of these demographic controls is interacted with postQE to allow for differential responses to rate declines. All regressions also include MSA and month fixed effects. For reference, the average monthly refinancing propensity over August 2008–January 2009 was 0.4%, while over February–July 2009 it was 1.2%. Descriptive statistics of explanatory variables are provided in Online Appendix Table A-1. MSAs are weighted by their 2008 population. Robust standard errors (clustered by MSA) are in parentheses. Significance: * $p < .1$, *** $p < .01$.

Read:

- High- and low-equity MSAs had similar refinancing trends before QE1.
- After QE1, refinancing rises much more in high-equity places.

- The cumulative gap reaches about 5 percentage points, while the cumulative refinancing propensity in the low-equity group is only about 7% over 2009.
- Table I shows that the equity interaction remains positive and highly significant after a long list of controls.

Economic message. The refinancing channel was active in 2008, but it was strongest exactly where borrower balance sheets were least damaged.

5.4 Figure IV: cash-out refinancing

Read:

- Cash extraction also rises more in high-equity MSAs after QE1.
- Across the top-equity MSAs, about $23.8 billion was cashed out in the six months after QE1.
- Across the bottom-equity MSAs, only about $10.9 billion was cashed out.
- After subtracting pre-QE trends, the differential increase is about $8 billion.

Economic message. The relevant margin is not just refinancing frequency. It is also how much liquidity refinancing creates. High-equity places both refinance more and extract more.

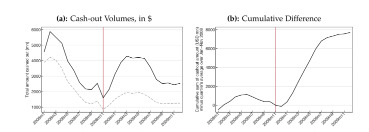

FIGURE IV

Cash-Out Refinancing in Top and Bottom Quartiles of MSAs by Median Borrower Equity in November 2008

Panel A shows total cash extracted during refinancing in the top and bottom MSA quartiles by $E^{med}_{j,Nov2008}$. Panel B shows the cumulative difference between the two groups, after subtracting each group's average cash-out amounts from January to November 2008. Since CRISM data does not cover the whole mortgage market, we scale up dollar amounts in CRISM for this figure; see Online Appendix A.2.3 for details. Vertical lines show the month of the QE1 announcement (November 2008).

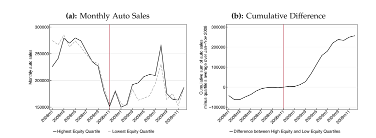

FIGURE V

New Auto Sales in Top and Bottom Quartiles of MSAs Defined by Median Borrower Equity in November 2008

Figure shows the total volume of new car purchases in the top and bottom MSA quartiles by $E^{med}_{j,Nov2008}$ using R.L. Polk data. Panel A shows total monthly car purchase volumes, summed over MSAs in each quartile. Panel B shows cumulative differences between the two groups, after subtracting from each group's series the group's average car purchases from January to November 2008. MSA groups are the same as in Figure III. The vertical line shows the month of QE1 announcement.

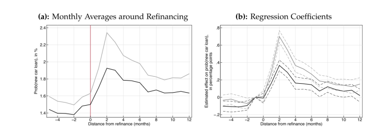

FIGURE VI

Borrower-Level Evidence on the Response of Car Purchase Propensity to Refinancing

Panel A shows average fractions of refinancing borrowers who obtain a new car loan in each month surrounding a mortgage refinancing (in month 0), distinguishing between cash-out and non-cash-out refinancing. We identify a borrower as obtaining a new car loan in month t if their car loan balance increases by $2,000 or more between month $t-1$ and month t. Panel B shows coefficients on month indicators (relative to the refinancing month, with -1 as the omitted month) from a regression of the new car loan indicator on calendar-month-by-MSA fixed effects as well as months-from-refinancing indicators interacted with whether the refinancing involves cash-out. Dashed lines are 95% confidence intervals based on robust standard errors clustered at the MSA level. See Online Appendix A.2.5 for additional details.

5.5 Figure V and Figure VI: spending responses

Read:

- New car sales diverge sharply after QE1 between high- and low-equity MSAs.
- The cumulative difference reaches roughly 250,000 cars by the end of 2009.
- At the household level, car-purchase propensities rise after refinancing, and the response is almost twice as large after cash-out refinancing as after non-cash-out refinancing.

Economic message. Refinancing affects real spending, and the cash-out margin is especially important.

5.6 Figure VII and Figure VIII: the channel changes over time

Read:

- Figure VII shows that mean house price growth, dispersion in house price growth, and the correlation between house prices and unemployment all vary substantially over time.

- During the Great Recession, house price growth and unemployment are unusually negatively correlated across states.
- Figure VIII shows that in 2008 refinancing is weakest in high-unemployment places, but in 2001 it is strongest there.

Economic message. The effect of lower rates depends on the collateral environment. The same easing cycle can have very different regional incidence in different recessions.

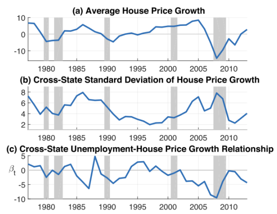

FIGURE VII

Changes in the Distribution of House Price Changes Across Time

Figure shows time series of the mean and standard deviation of state-level annual house price growth and of the cross-state relationship between house price growth and unemployment rate changes. Calculations are population weighted by state.

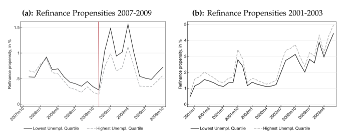

FIGURE VIII

Refinancing Propensities for High and Low Unemployment Change Quartiles, 2008 Recession and 2001 Recession

Figure shows monthly refinancing propensities for the top and bottom quartiles of MSAs by unemployment changes. Specifically, for each MSA we compute the change in unemployment between November 2007 and October 2009 (Panel A) and December 2000 to June 2003 (Panel B). Each MSA is placed into population-weighted quartiles by this measure. Refinancing propensities are computed using HMDA data on origination volumes (by month of loan application) divided by the total number of mortgages in MSA quartiles from the American Community Survey in 2008 (Panel A) or the Census in 2000 (Panel B). The vertical line in Panel A shows the month of QE1 announcement.

TABLE II

AGGREGATE REFINANCING PROPENSITIES AND INTEREST RATES

Dep. var.:	$Refi_t$		$\Delta Refi_{t-12,t}$			
Int. rate measure:	Mtg. rate minus avg. outstanding		$\Delta Mtg.rate_{t-12,t}$		Σ_{t-12}^{t}Mon. pol. shocks	
	(1)	(2)	(3)	(4)	(5)	(6)
Rate	−0.697***	−0.962***	−0.977***	−0.983***	−1.387***	−2.094***
	(0.181)	(0.139)	(0.126)	(0.128)	(0.478)	(0.532)
ΔHP		0.063***		0.004		0.007
		(0.012)		(0.021)		(0.029)
Rate × ΔHP		−0.095***		−0.070**		−0.248**
		(0.030)		(0.032)		(0.120)
SD(ΔHP)		0.116***		−0.076		−0.106
		(0.036)		(0.060)		(0.084)
Rate × SD(ΔHP)		−0.178**		−0.176*		−0.944*
		(0.077)		(0.092)		(0.490)
Constant	1.003***	0.964***	−0.241**	−0.180*	−0.192	−0.193
	(0.077)	(0.060)	(0.113)	(0.108)	(0.148)	(0.124)
Adj. R^2	0.22	0.63	0.50	0.56	0.12	0.20
Obs.	222	222	258	258	258	258

Notes. Table shows results from regressions of monthly aggregate refinancing propensities on measures of (mortgage) interest rates. In columns (1) and (2), the interest rate measure is the mortgage rate on new 30-year FRMs (from Freddie Mac PMMS) minus the weighted-average coupon on outstanding mortgages guaranteed by Fannie Mae, Freddie Mac, and Ginnie Mae (from eMBS, available since January 1994). In columns (3) and (4), the interest rate variable is the 12-month change in the rate on new 30-year FRMs (one standard deviation = 0.78). In columns (5) and (6), the interest rate variable is the 12-month sum of monthly monetary policy surprises (the surprise in the three-month ahead futures rate) from Gertler and Karadi (2015) (one standard deviation = 0.28). Sample period: columns (1)–(2): January 1994–June 2012; (3)–(6): Jan 1991–June 2012. June 2012 is when the Gertler-Karadi series ends. Standard errors in parentheses are Newey-West (4 lags) in columns (1)–(2) and Hansen-Hodrick (12 lags) in columns (3)–(6). Significance: $^{*}p < .1$, $^{**}p < .05$, $^{***}p < .01$.

5.7 Table II: aggregate refinancing across time

Read:

- The direct rate effect is always negative: lower mortgage rates increase refinancing.
- The interaction Rate × ΔHP is negative and significant.
- The interaction Rate × $SD(\Delta HP)$ is also negative and significant in the key specifications.

Economic message. Aggregate refinancing responds more strongly to lower rates when average house price growth is higher and when the distribution of house price growth is more favorable for refinancing.

5.8 Figures IX and X: nonlinear refinancing policy and regional model fit

Read:

- Figure IX shows that refinancing follows a threshold rule in extractable equity.
- Figure X shows that the calibrated model reproduces the empirical pattern that high-equity regions respond more to the same rate decline.

Economic message. The model matches the regional evidence because it builds in the correct microeconomic nonlinearity: underwater borrowers do not refinance.

5.9 Figure XI and Figure XII: why 2008 differs from 2001

Read:

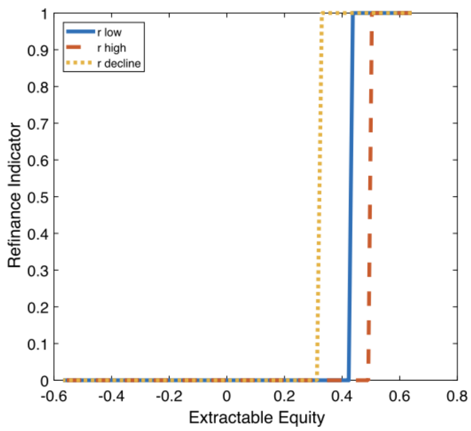

FIGURE IX

Refinance Decision Follows a Threshold Policy

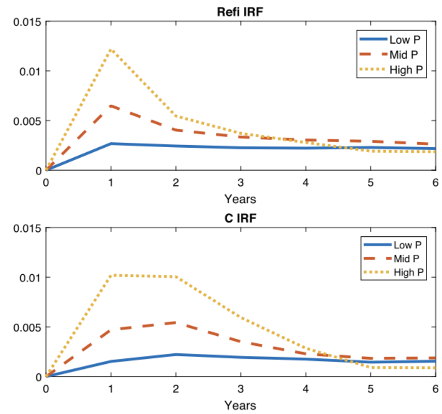

FIGURE X

Response to Interest Decline by Region

For the baseline 2008 calibration, the refinancing impulse response function (IRF) shows the change in the monthly fraction of households refinancing in response to a one percentage point reduction in mortgage rates. The consumption (C) IRF shows the change in log consumption in response to the same one percentage point reduction in mortgage rates.

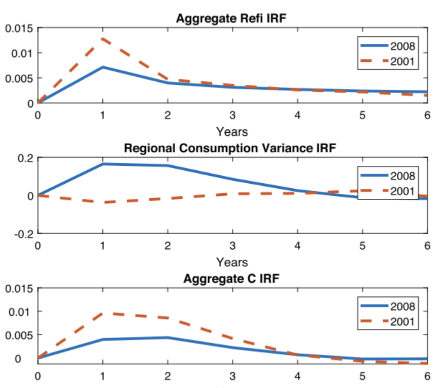

FIGURE XI

Aggregate Stimulus versus Cross-regional Inequality in 2008 and 2001

Impulse response functions show the change in the monthly fraction of households refinancing, log aggregate consumption, and log consumption variance across regions in response to a one percentage point reduction in mortgage rates. See text for the description of the baseline economy calibrated to match November 2008 as well as the model specification for the 2001 recession.

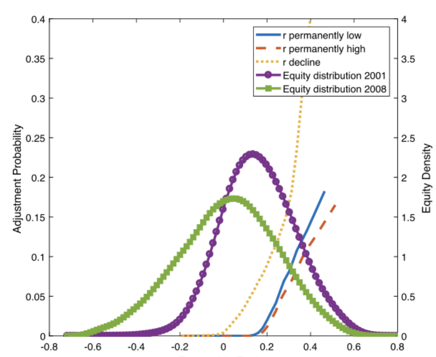

FIGURE XII

Distribution of Equity and Refinancing Probability: 2008 Calibration versus 2001

This figure shows the simulated distribution of equity and the fraction of households refinancing under two different equity distributions.

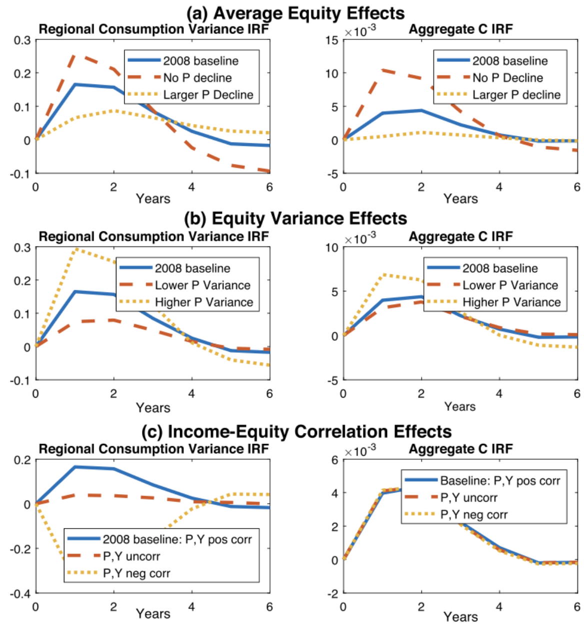

FIGURE XIII

Effects of Changing Equity Distribution

Impulse response functions show the change in log aggregate consumption and in log consumption variance across regions in response to a one percentage point reduction in mortgage rates. The baseline economy includes a 12.5% aggregate house price decline. In Panel A, the "larger P decline" calibration features a 25% decline in house prices. The variance of equity and its correlation with income are fixed at the 2008 calibration across all simulations. In Panel B, the high variance calibration doubles the difference between high and low house price regions, while the low variance calibration halves it. All economies feature the same baseline decline in house prices and correlation with income. In the bottom panel, the baseline calibration has income and house prices positively correlated across regions. In the other two calibrations, they are uncorrelated or negatively correlated. All simulations feature the same baseline decline in house prices and variance across regions.

- Under 2008 conditions, the same one percentage point mortgage-rate decline raises aggregate consumption by about 0.41%.
- Under 2001-style conditions, the same shock raises aggregate consumption by about 0.92%, or about 2.25 times as much.
- Figure XII shows why: the 2008 equity distribution is shifted left and more dispersed, leaving less mass in the part of the equity distribution where refinancing responds strongly.

Economic message. The weak aggregate effect of QE1 in 2008 is not because lower rates stopped mattering. It is because the equity distribution moved into a region where many households could not use them.

5.10 Figure XIII: which moments of the equity distribution matter

Read:

- Raising mean equity increases the aggregate consumption response to a rate cut.
- Raising the variance of equity can also amplify the response because it puts more mass in high-equity regions, though the effect is nonlinear.

- The correlation between income and equity mostly governs whether rate cuts raise or reduce cross-regional inequality.

Economic message. Looking only at the mean of the equity distribution is not enough. Different moments of the distribution govern stimulus and inequality through different channels.

5.11 Figure XIV: mortgage modification policies

Read:

- Debt forgiveness by itself raises consumption and reduces inequality.
- Relaxing refinancing constraints by itself does little if rates do not move.
- When rates fall, both debt forgiveness and relaxed refinancing requirements amplify stimulus and reduce inequality relative to the baseline monetary-policy-only case.

Economic message. Fiscal or regulatory mortgage interventions can complement monetary policy when the refinancing channel is blocked by negative equity.

6 Short summary

- The paper studies how the regional distribution of home equity changes the refinancing channel of monetary policy.
- Using mortgage microdata, it shows that QE1 induced much more refinancing, cash-out, and spending in high-equity regions than in low-equity regions.
- The same channel looks very different in 2001, when house prices were not collapsing and high unemployment was not tied to low housing equity.
- Aggregate refinancing responds more to lower rates when mean house price growth is higher and when the cross-regional distribution of house price growth is more favorable.
- A heterogeneous-household model with refinancing frictions and lenders implies that under 2008 conditions monetary easing provided weak aggregate stimulus and increased regional consumption inequality.
- Under 2001-style collateral conditions, the same rate cut would have produced much larger stimulus.
- The key mechanism is the nonlinear relation between refinancing and extractable equity.
- Mortgage modification policies that help underwater borrowers refinance can restore the potency of monetary easing.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that the refinancing channel of monetary policy is highly state dependent because it depends on the *distribution of home equity*. In the Great Recession, the regional pattern of collateral destruction prevented lower mortgage rates from stimulating the regions that needed it most. This dampened aggregate stimulus and increased cross-regional consumption inequality.

7.2 Economic intuition

Monetary easing through mortgages is not a smooth linear mechanism. It operates through a constraint. If a household has enough equity, a lower mortgage rate can trigger refinancing, lower payments, and release cash. If the household is underwater, the same interest-rate decline does little. Therefore, a leftward shift in the equity distribution can sharply weaken monetary transmission even if rates fall by the same amount.

7.3 Takeaway

The paper's broad lesson is that collateral distributions are macroeconomically important. Central banks and policymakers should not think about housing merely through average prices or average leverage. What matters for the transmission of policy is *who has equity, where they are located, and how that interacts with local economic distress*. In that sense, the distribution of housing equity is a state variable for monetary policy.